Solved Exercise 4.1

1. Factorize by identifying common factors.

(i) $6x+12=6(x+2)$.
(ii) $15y^2+20y=5y(3y+4)$.
(iii) $-12x^2-3x=-3x(4x+1)$.
(iv) $4a^2b+8ab^2=4ab(a+2b)$.
(v) $xy-3x^2+2x=x(y-3x+2)$.
(vi) $3a^2b-9ab^2+15ab=3ab(a-3b+5)$.

2. Factorize.

(i) $5x+15=5(x+3)$.
(ii) $x^2+4x+3 = x^2+x+3x+3$
$= x(x+1)+3(x+1)$
$= (x+1)(x+3)$.
(iii) $x^2+6x+8 = x^2+2x+4x+8$
$= x(x+2)+4(x+2)$
$= (x+2)(x+4)$.
(iv) $x^2+4x+4 = x^2+2(x)(2)+2^2$
$= (x+2)^2$.

3. Factorize the eight trinomials by splitting the middle term.

(i) $x^2+x-12 = x^2+4x-3x-12$
$= x(x+4)-3(x+4)$
$= (x+4)(x-3)$.
(ii) $x^2+7x+10 = x^2+5x+2x+10$
$= x(x+5)+2(x+5)$
$= (x+5)(x+2)$.
(iii) $x^2-6x+8 = x^2-4x-2x+8$
$= x(x-4)-2(x-4)$
$= (x-4)(x-2)$.
(iv) $x^2-x-56 = x^2-8x+7x-56$
$= x(x-8)+7(x-8)$
$= (x-8)(x+7)$.
(v) $x^2-10x-24 = x^2-12x+2x-24$
$= x(x-12)+2(x-12)$
$= (x-12)(x+2)$.
(vi) $y^2+4y-12 = y^2+6y-2y-12$
$= y(y+6)-2(y+6)$
$= (y+6)(y-2)$.
(vii) $y^2+13y+36 = y^2+9y+4y+36$
$= y(y+9)+4(y+9)$
$= (y+9)(y+4)$.
(viii) $x^2-x-2 = x^2-2x+x-2$
$= x(x-2)+(x-2)$
$= (x-2)(x+1)$.

4. Factorize each quadratic.

(i) $2x^2+7x+3 = 2x^2+6x+x+3$
$= 2x(x+3)+(x+3)$
$= (2x+1)(x+3)$.
(ii) $2x^2+11x+15 = 2x^2+6x+5x+15$
$= 2x(x+3)+5(x+3)$
$= (2x+5)(x+3)$.
(iii) $4x^2+13x+3 = 4x^2+12x+x+3$
$= 4x(x+3)+(x+3)$
$= (4x+1)(x+3)$.
(iv) $3x^2+5x+2 = 3x^2+3x+2x+2$
$= 3x(x+1)+2(x+1)$
$= (3x+2)(x+1)$.
(v) $3y^2-11y+6 = 3y^2-9y-2y+6$
$= 3y(y-3)-2(y-3)$
$= (3y-2)(y-3)$.
(vi) $2y^2-5y+2 = 2y^2-4y-y+2$

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$= 2y(y-2)-(y-2)$
$= (2y-1)(y-2).$
(vii) $4z^2-11z+6 = 4z^2-8z-3z+6$
$= 4z(z-2)-3(z-2)$
$= (4z-3)(z-2).$
(viii) $6+7x-3x^2 = 6+9x-2x-3x^2$
$= 3(2+3x)-x(2+3x)$
$= (3-x)(3x+2).$

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